



C2401 刷机教程

Jetson Orin Nano / Orin NX • SDK Manager • SUPER 固件

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1. C2401 刷机

提示：不同载板、核心卡和 JetPack 版本的刷机参数不同，请按对应章节操作。

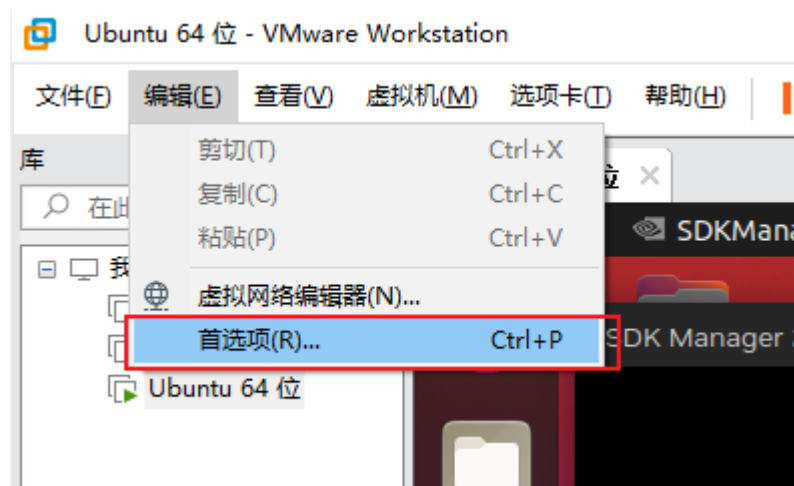
C2401 刷入系统

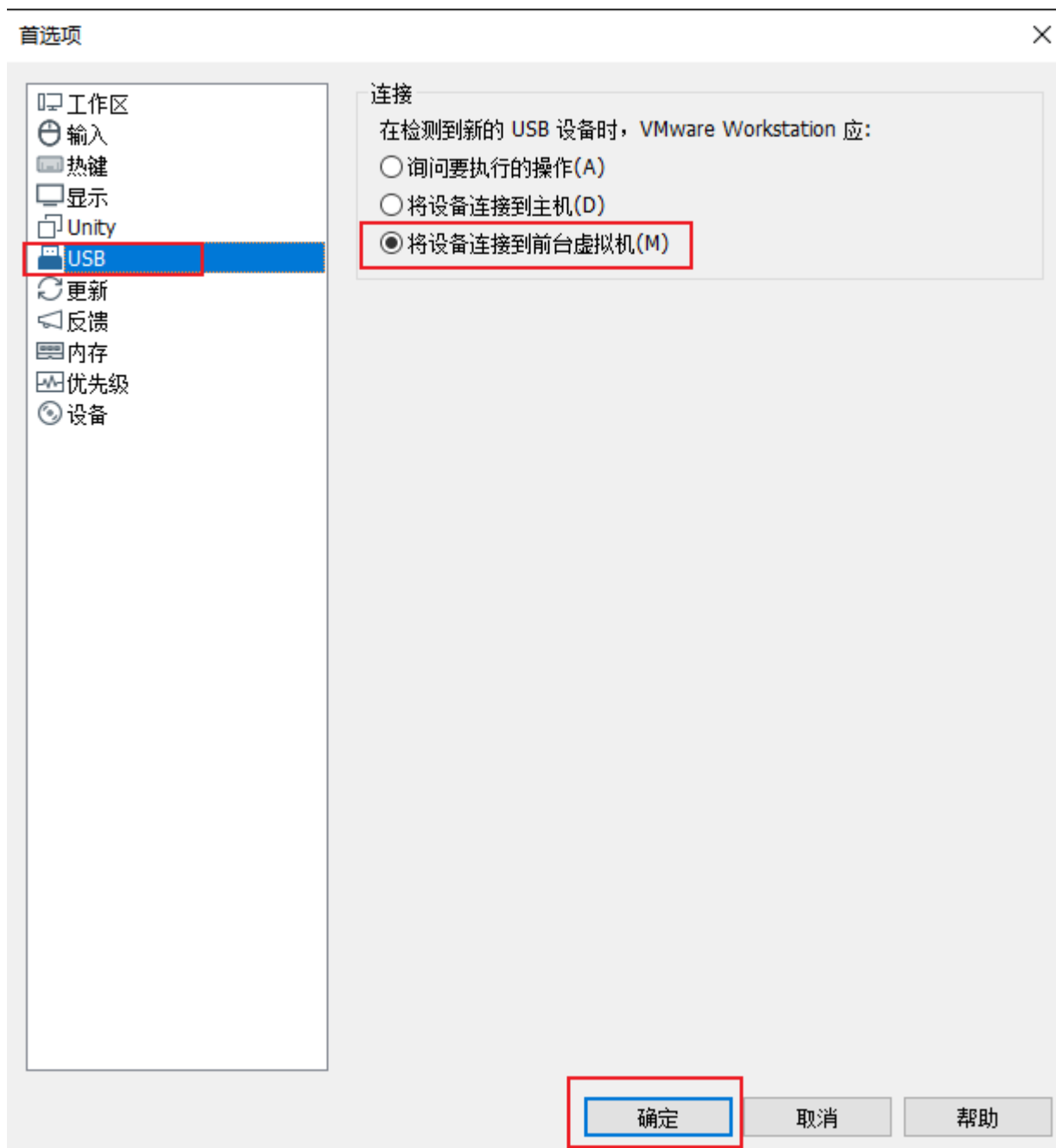
建议在 Linux 系统或 Linux 虚拟机下进行刷机操作，虚拟机和 SDK Manager 安装方法可参考 [VMware 虚拟机安装说明](#)。

1. 使用 NVIDIA SDK Manager 进行刷机

1.1 设置 VMware 的 USB 模式

将 USB 连接模式设置为「将设备连接到前台虚拟机」，避免刷机中途开发板重启导致失败。





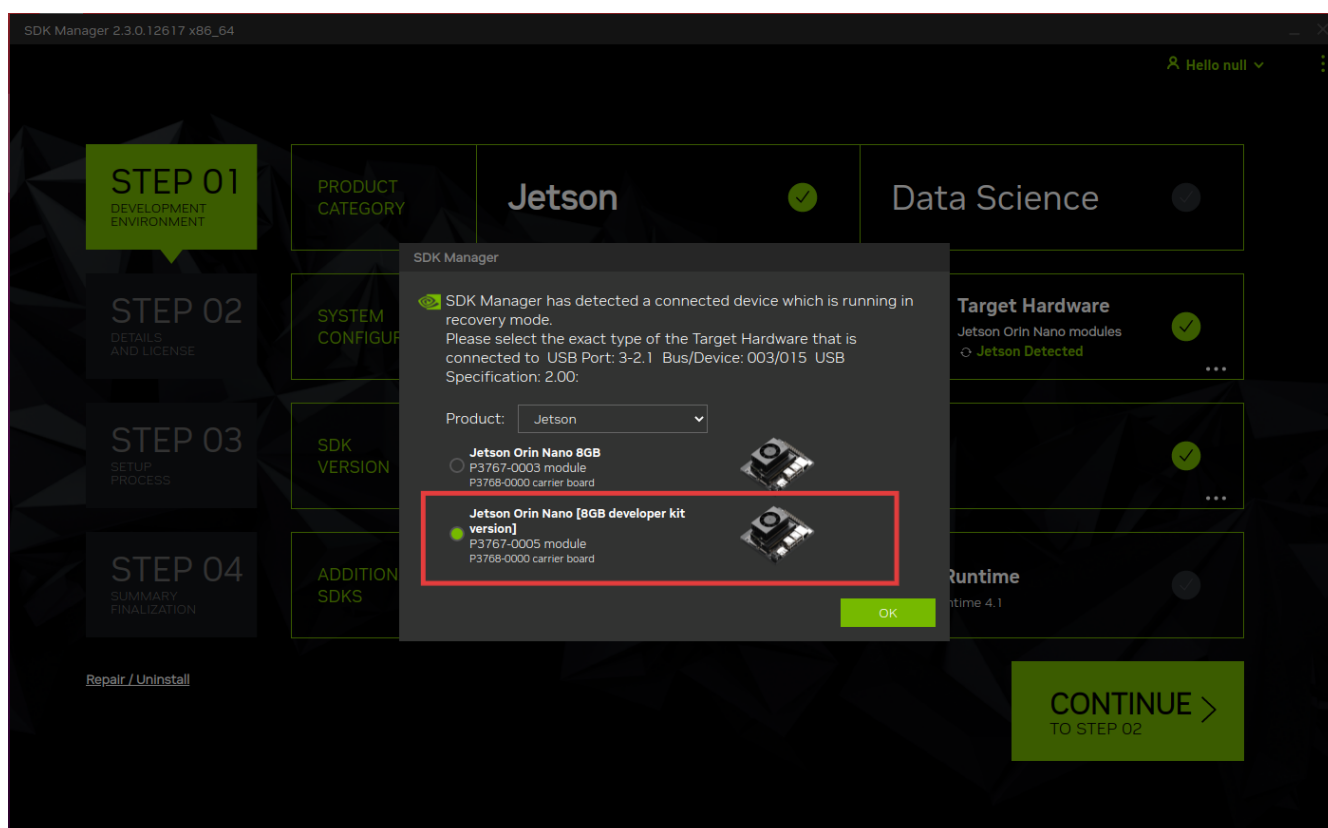
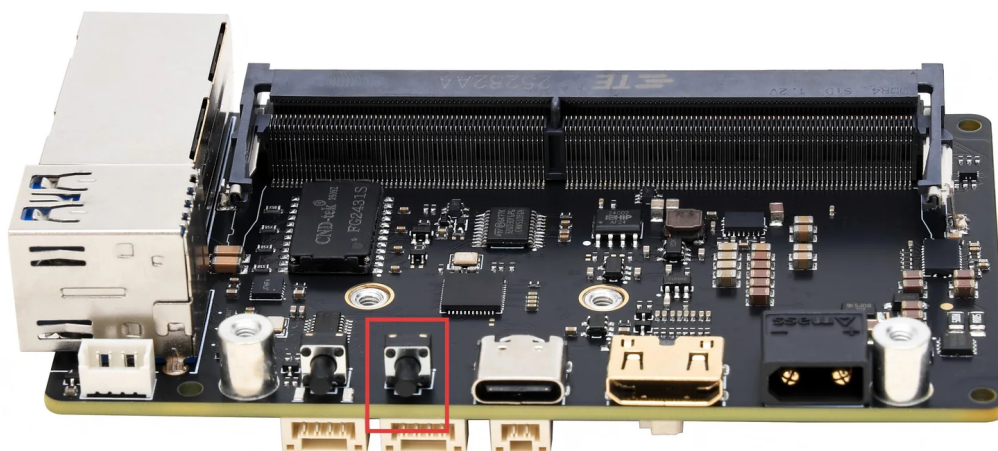
1.2 进入恢复模式

C2401 有两种方式进入恢复模式：

方式一（按键法）： 在断电的情况下，按住 **REC 按钮**不放，再接通电源，即可进入恢复模式。

方式二（拨码法）： 将载板背面的**拨码开关 3** 拨动至 ON，每次上电时系统都将自动进入恢复模式。

恢复模式仅持续 60 秒，系统检测到无刷机操作后会自动进入正常开机时序。



1.3 配置 SDK

- 取消 Host Machine，并点击 CONTINUE

- 只勾选 Jetson Linux

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STEP 02
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SUMMARY
FINALIZATION

PRODUCT CATEGORY	Jetson ✓	Data Science ✓
SYSTEM CONFIGURATION	Host Machine Ubuntu 20.04 - x86_64 ✓	Target Hardware Jetson Orin Nano modules ✓ Jetson Orin Nano 8GB 1 ...
SDK VERSION	JetPack 6.2 ✓ See what's new ...	
ADDITIONAL SDKS	DeepStream DeepStream 7.1 ✓	GXF Runtime GXF Runtime 4.1 ✓

Repair / Uninstall

CONTINUE >
TO STEP 02

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JETPACK 6.2.1 LINUX FOR JETSON ORIN NX MODULES [Expand all](#)

✓ TARGET COMPONENTS	DOWNLOAD SIZE	STATUS
✓ Jetson Linux ✓ Jetson Linux image 2,458 MB OS image ready ✓ Flash Jetson Linux		
Jetson Runtime Components Additional Setups CUDA Runtime CUDA X-AI Runtime Computer Vision Runtime NVIDIA Container Runtime Multimedia		
Jetson SDK Components CUDA CUDA-X AI		

System requires up to 10GB (host) and 0GB (target) of available disk space during setup.

Download folder: /home/ubuntu/Downloads/nvidia/sdkm_downloads [change](#)

SDKs install folder: /home/ubuntu/nvidia/nvidia_sdk [change](#) (10GB required)

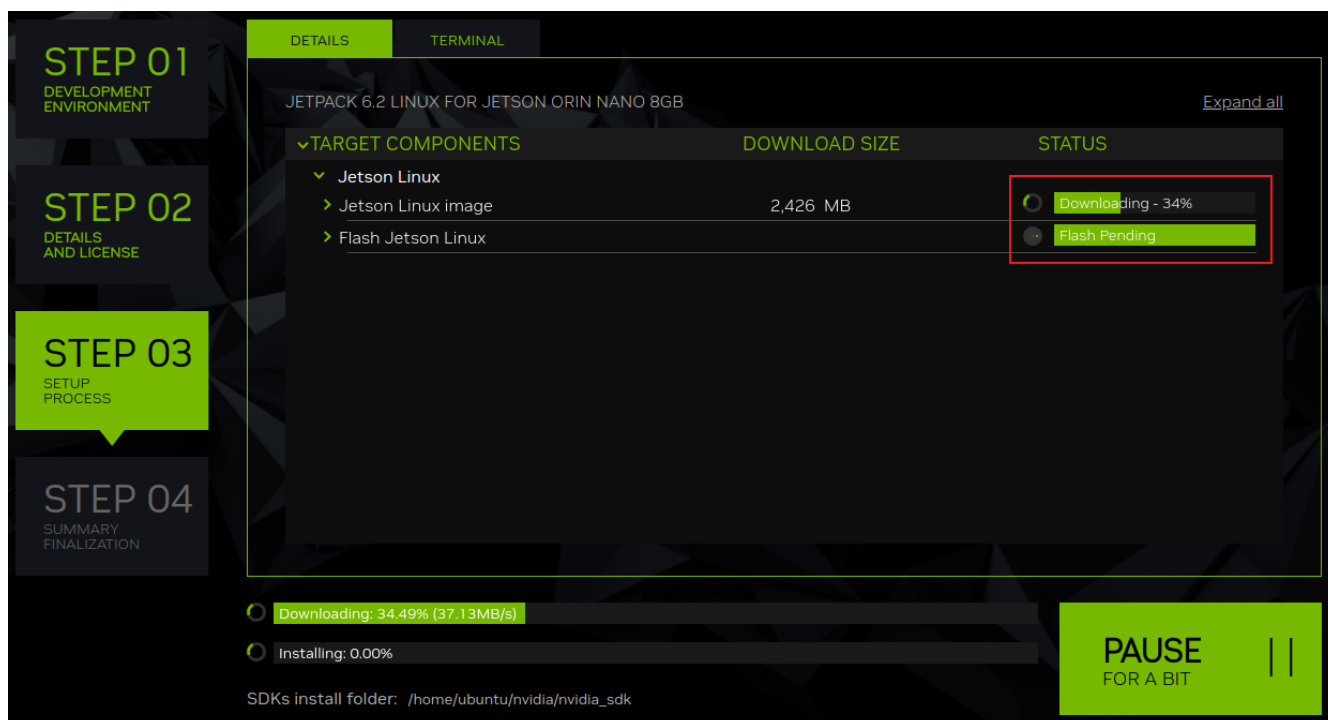
☒ I accept the terms and conditions of the [license agreements](#).

☐ Download now. Install later.

CONTINUE >
TO STEP 03

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1.4 等待下载完成

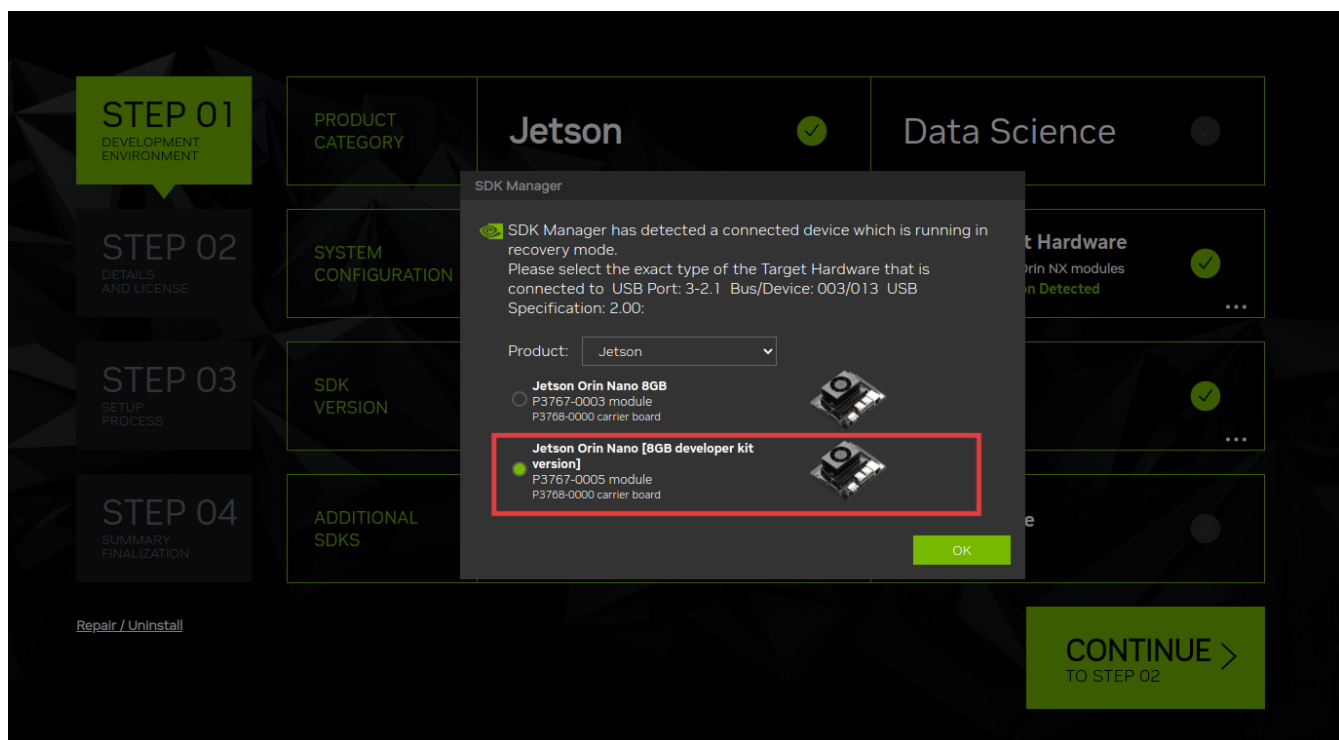
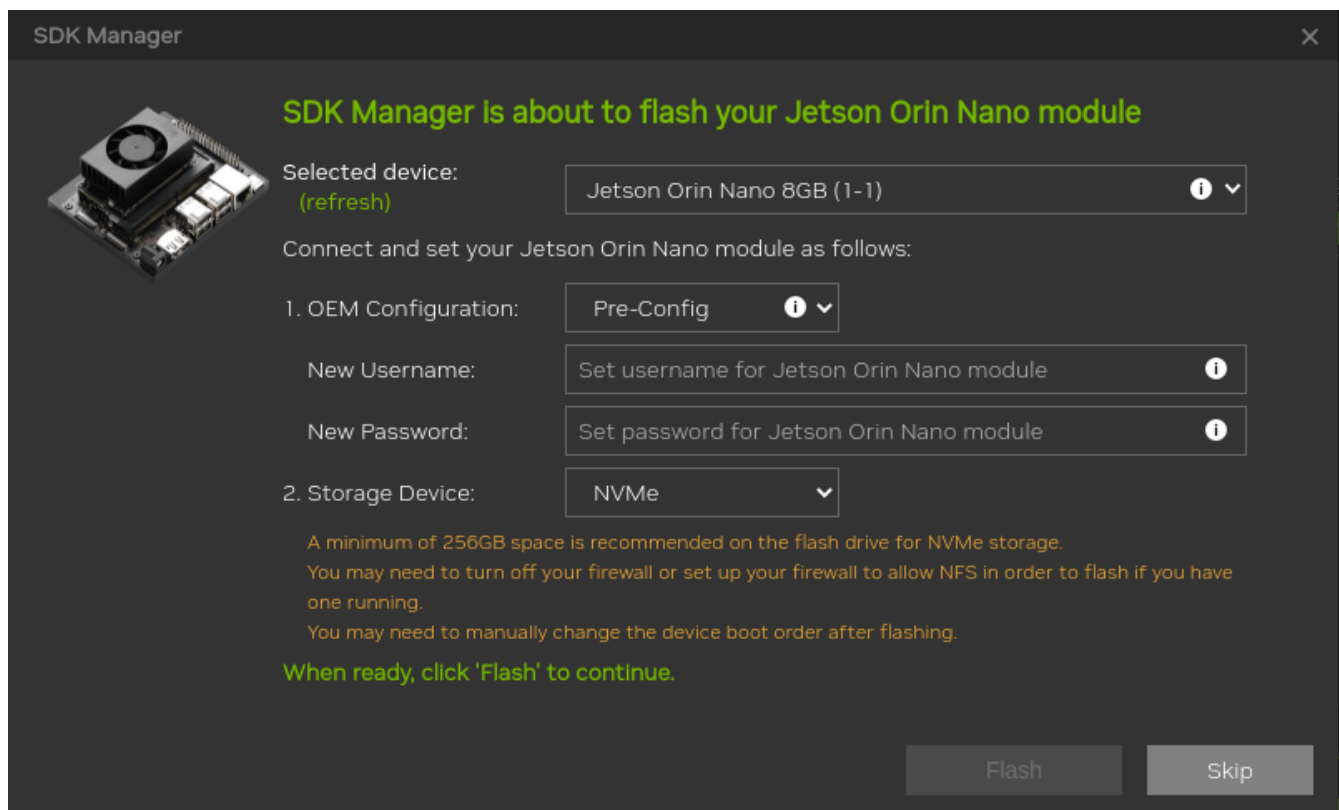


1.5 配置刷机参数

- **Pre-Config**: 预设账号密码
- **Runtime**: 开机后自行配置账号密码
- **Storage Device**: 系统刷入的介质（固态硬盘）
- 选择 **Developer Kit Version**

耐心等待刷机完成，大约 10~20 分钟。刷入过程中开发板可能会多次与主机重新连接，请勿途中拔开数据线或切断开发板电源。

刷入完成后，上电正常进入系统桌面或初始化设置页面即刷机成功。



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DETAILS

TERMINAL

JETPACK 6.2 LINUX FOR JETSON ORIN NANO 8GB

Expand all

✓ TARGET COMPONENTS	DOWNLOAD SIZE	STATUS
✓ Jetson Linux		
➤ Jetson Linux image	2,426 MB	✓ OS image ready
➤ Flash Jetson Linux		✓ Success

✓ INSTALLATION COMPLETED SUCCESSFULLY.

EXPORT LOGS

FINISH
AND EXIT

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SDK Manager 2.3.0.12617 x86_64

Hello null

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DETAILS

TERMINAL

Component Name: All

Message Type: All

Filter text here...

Find

15:59:54 INFO: command finished successfully

16:01:04 DEBUG: running command < true >

16:01:04 INFO: command finished successfully

16:01:04 DEBUG: running command < lsusb | grep 0955: | awk '{printf \$2 "/" \$4 " " \$6 ";"}' >

16:01:04 INFO: command finished successfully

16:01:10 DEBUG: running command < true >

16:01:10 INFO: command finished successfully

16:01:10 DEBUG: running command < lsusb | grep 0955: | awk '{printf \$2 "/" \$4 " " \$6 ";"}' >

16:01:10 INFO: 001/021: 0955:7020;

16:01:10 INFO: command finished successfully

16:01:37 INFO: Flash Jetson Linux - flash: done.

16:01:37 INFO: Flash Jetson Linux - flash: Exiting with code 0

16:01:37 INFO: Flash Jetson Linux - flash: [Component Install Finished Successfully]

16:01:37 INFO: Flash Jetson Linux - flash: [host] [768.00 KB released. Disk Avail on Partition /dev/sda5: 198.96 GB]

16:01:37 INFO: Flash Jetson Linux - flash: [NV_L4T_FLASH_JETSON_LINUX_COMP Install took 17m53s]

16:01:37 INFO: Flash Jetson Linux - flash: command finished successfully

16:01:37 SUMMARY: Flash Jetson Linux - flash: Install completed successfully.

16:01:37 INFO: All done!

✓ INSTALLATION COMPLETED SUCCESSFULLY.

EXPORT LOGS

FINISH
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2. 使用命令行刷入 SUPER 固件

此操作需依赖官方固件环境，请先通过 SDK Manager 完成至少一次完整的烧录以创建固件缓存。

2.1 进入恢复模式

按住 REC 按钮，使用 Type-C 数据线连接至电脑并接通开发板电源，使开发板进入恢复模式。

2.2 使用官方固件进行命令行刷机

操作前请关闭正在运行的 SDK Manager 软件，刷入过程约为 10~20 分钟。

JetPack 6.2 版本刷入指令：

```
cd /home/ubuntu/nvidia/nvidia_sdk/JetPack_6.2.1_Linux_JETSON_ORIN_NANO_TARGETS/Linux_for_Tegra
sudo ./tools/kernel_flash/l4t_initrd_flash.sh --external-device nvme0n1p1 \
  -c tools/kernel_flash/flash_l4t_t234_nvme.xml -p "-c bootloader/generic/cfg/flash_t234_qspi.xml" \
  --showlogs --network usb0 jetson-orin-nano-devkit-super internal
```

JetPack 5.1.5 版本刷入指令：

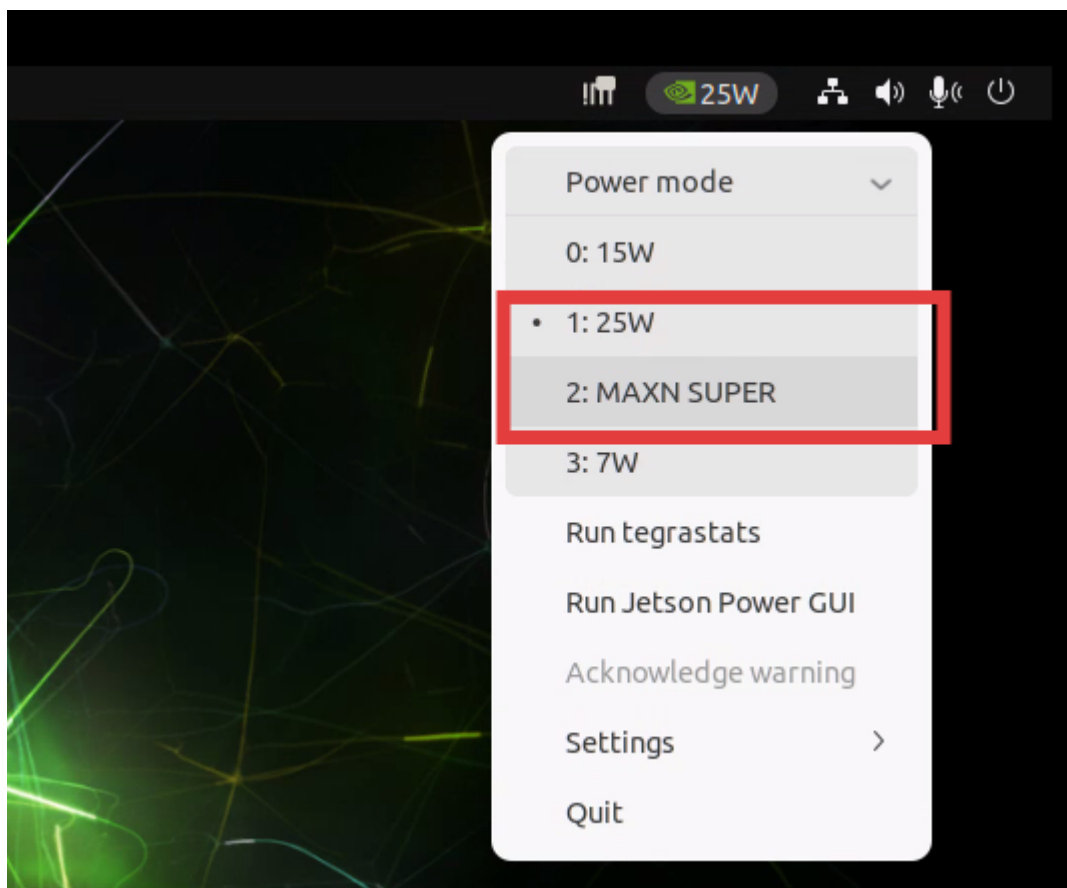
```
cd /home/ubuntu/nvidia/nvidia_sdk/JetPack_5.1.5_Linux_JETSON_ORIN_NANO_TARGETS/Linux_for_Tegra
sudo ./tools/kernel_flash/l4t_initrd_flash.sh --external-device nvme0n1p1 \
-c tools/kernel_flash/flash_l4t_external.xml -p "-c bootloader/t186ref/cfg/flash_t234_qspi.xml" \
--showlogs --network usb0 jetson-orin-nano-devkit-super internal
```



```
ubuntu@ubuntu-virtual-machine: ~/nvidia/nvidia_sdk/JetPac...
00:03:33.134 - Debug: Sha1 checksum matched for /mnt/internal/qspi_bootblob_ver.txt
00:03:33.137 - Debug: Writing /mnt/internal/qspi_bootblob_ver.txt (109 bytes) in to /dev/mtd0:66977792
Copied 109 bytes from /mnt/internal/qspi_bootblob_ver.txt to address 0x03fe0000 in flash
00:03:33.164 - Debug: Writing gpt_secondary_3_0.bin (parittion: secondary_gpt) into /dev/mtd0
00:03:33.182 - Debug: Sha1 checksum matched for /mnt/internal/gpt_secondary_3_0.bin
00:03:33.185 - Debug: Writing /mnt/internal/gpt_secondary_3_0.bin (16896 bytes) into /dev/mtd0:67091968
Copied 16896 bytes from /mnt/internal/gpt_secondary_3_0.bin to address 0x03ffbe00 in flash
00:03:33.219 - Info: Successfully flashed the QSPI.
00:03:33.222 - Info: Flashing success
00:03:33.244 - Debug: The device size indicated in the partition layout xml is smaller than the actual size. This utility will try to fix the GPT.
Flash is successful
Reboot device
Cleaning up...
Log is saved to Linux_for_Tegra/initrdlog/flash_1-1_0_20250709-164603.log
ubuntu@ubuntu-virtual-machine:~/nvidia/nvidia_sdk/JetPack_6.2.1_Linux_JETSON_ORIN_NANO_TARGETS/Linux_for_Tegra$
```

2.3 验证是否处于 SUPER 模式

开发板上电开机完成用户配置后，在桌面右上角可以选择电源模式。**25W & MAXN SUPER** 为 SUPER 模式独有，普通模式只有 7W 和 15W 两档。



3. 备份与恢复现有固件

3.1 备份固件

按住 REC 按钮，Type-C 连接电脑，接通电源进入恢复模式。

或在正常开机状态下连接到主机，输入以下命令重启到恢复模式：

```
sudo reboot -f forced-recovery
```

进入原刷机固件缓存下的命令行目录：

```
cd /home/ubuntu/nvidia/nvidia_sdk/JetPack_6.2.1_Linux_JETSON_ORIN_NANO_TARGETS/Linux_for_Tegra
```

安装依赖包：

```
sudo apt-get install qemu-user-static libxml2-utils abootimg sshpass nfs-kernel-server binutils
```

备份命令：

```
sudo ./tools/backup_restore/l4t_backup_restore.sh -b -e nvme0n1 jetson-orin-nano-devkit-nvme
```

恢复命令：

```
sudo ./tools/backup_restore/l4t_backup_restore.sh -r -e nvme0n1 jetson-orin-nano-devkit-nvme
```

若使用第三方载板恢复备份，需要修改 `nvrestore_partitions.sh` 注释掉 292~296 行的内容。

```
278 # This block will check to make sure the model of the target board matches the
279 # model of the board the images came from.
280 for value in $(grep -v -e '(\^ *$|^#)' < "${FILE_NAME}"); do
281     declare -a FIELDS
282     for part in {1..6}; do
283         FIELDS[part]=$(echo "$value" | awk -F, -v part=${part} '{print $part}')
284     done
285     if [ "${FIELDS[1]}" = 'board_spec' ]; then
286         set_board_spec
287         if [[ "${FIELDS[2]}" == "${BOARD_SPEC}" ]]; then
288             BOARD_MATCH=true
289         fi
290     fi
291 done
292 if [ ${BOARD_MATCH} = false ]; then
293 #     echo "${SCRIPT_NAME}: You are trying to flash images from a board model that does not"
294 #     echo "match the current board you're flashing onto."
295 #     exit 1
296 #fi
297
298 declare -A able_to_delete
```